VINNA: Verifiable Intelligence for Navigating and Analyzing Construction Sites from Egocentric Video

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Abstract

Construction sites are spatially complex, dynamic environments where workers, heavy equipment, materials, and structural elements continuously reconfigure across three dimensions. Understanding this spatial evolution—distances between entities, zone occupancy patterns, temporal changes in site layout—is a prerequisite for every downstream task in construction intelligence, from safety compliance to progress monitoring to resource allocation. Yet current vision systems for construction treat spatial reasoning as an afterthought, jumping directly to narrow classification tasks (PPE detection, violation flagging) without building a general spatial representation.

We introduce VINNA (**V**erifiable **I**ntelligence for **N**avigating and **A**nalyzing construction sites), a framework for egocentric spatial reasoning on construction footage. VINNA extracts metric distance estimates, spatial relationship graphs, zone activity classifications, and temporal change signals from body-worn camera and LiDAR streams. Crucially, each spatial claim is paired with a *verifiable reward*: distance estimates are checked against fused depth data, spatial relationships are validated by 3D geometry, and change detection outputs are scored against frame-pair differencing. This verifiability enables self-supervised training via GRPO without human annotation.

On 638 frames (21 minutes) of Ironsite egocentric fisheye + LiDAR footage from an active masonry site, organized into 39 temporal batches with 3072-dimensional Gemini Embedding 2 representations, VINNA produces spatially grounded scene analyses that cluster into semantically meaningful groups (within-cluster cosine similarities of 0.4246, 0.4127, and 0.3765 across three spatial condition tiers). As a downstream demonstration, we show that VINNA's spatial representations enable CII compliance classification (P/C/NC) at 84% accuracy on a 25-frame annotated subset—without any compliance-specific training.

This paper is an honest report of a 48-hour hackathon prototype with real construction data, not a polished conference submission. We describe what works, what does not, and what the spatial intelligence gap in construction AI looks like from the ground up.

Keywords

spatial intelligence, egocentric video, construction, distance estimation, change detection, verifiable rewards, VLM, GRPO, LiDAR, spatial reasoning

1 Introduction

Construction is a $13 trillion global industry where the physical environment changes daily [9]. Every decision on a construction site—where to place materials, whether a scaffold is safe, when to pour concrete, how to route equipment—is fundamentally a *spatial* decision. Yet the computer vision systems deployed on construction sites today largely bypass spatial reasoning entirely.

The dominant paradigm is narrow classification: detect hard hats, flag missing guardrails, count workers in a zone [10, 11]. These systems answer "what is present?" but not "where is it relative to everything else, and how has that changed?" The spatial questions—*How far is the worker from the edge? Has the scaffold been moved since yesterday? Which zone has the highest activity density this hour?*—are the ones that actually drive operational decisions.

This gap between spatial questions and classification answers is not unique to construction. The broader vision-language model (VLM) community has recently recognized *spatial intelligence* as a first-class research problem. SpatialVLM [1] demonstrated that VLMs can learn metric distance estimation from synthetic spatial QA data generated via monocular depth. SpatialRGPT [2] extended this to region-grounded spatial queries over 3D scene graphs. Ego4D [5] established egocentric video understanding as a research domain, though its construction coverage is limited to activity recognition rather than spatial analysis.

None of these systems target construction, and none address the specific spatial challenges of the domain: extreme fisheye distortion, LiDAR occlusion from equipment, rapid scene change from active work, and the need for *metric* (not just relative) spatial claims grounded in site coordinates.

Meanwhile, the reinforcement learning community has converged on *verifiable rewards* as a superior alternative to human preference labels for training language models on tasks with checkable answers [3]. Spatial reasoning is precisely such a task: if a model claims "the worker is 2.3 meters from the scaffold edge," we can verify that claim against the fused LiDAR depth data. Smooth Operator [4] showed that continuous sigmoid-based rewards (SNRA) provide stable training gradients for numerical spatial predictions—exactly the reward shape we need for distance estimation.

The VINNA thesis. Construction sites need *spatial intelligence*, not violation classifiers. Spatial understanding—distance estimation, change detection, spatial relationship mapping, zone activity analysis—is the foundation. Compliance classification, progress monitoring, resource optimization, and safety alerting are all *downstream applications* of that foundation. A system that understands space can do all of these; a system that only detects hard hats can do one.

Contributions.

(1) We frame construction-site computer vision as a *spatial intelligence* problem and identify four core spatial capabilities:

metric distance estimation, spatial relationship extraction, temporal change detection, and zone activity analysis.

(2) We design verifiable reward functions for each spatial capability—distance rewards checked against LiDAR depth, relationship rewards validated by 3D geometry, change detection rewards scored by frame-pair differencing—enabling self-supervised GRPO training without human labels.

(3) We demonstrate the approach on 638 frames of real Ironsite egocentric footage, showing that spatial representations produce meaningful embedding-space structure (39 temporal batches, 3072-dim embeddings).

(4) We show CII compliance classification as a downstream task that emerges from spatial understanding, achieving 84% accuracy without compliance-specific training.

Scope and honesty. VINNA is a 48-hour hackathon prototype. We have real data, real embeddings, and a real pipeline. We do not have a trained RL model—the GRPO training loop is designed but not yet executed. We report what we built, what the data shows, and what the spatial intelligence framing makes possible.

2 Related Work

2.1 Spatial Reasoning in Vision–Language Models

The spatial intelligence gap in VLMs has attracted significant recent attention.

SpatialVLM [1] generates 2 billion spatial QA pairs from monocular depth estimation, training VLMs to answer metric distance and spatial relationship questions. Their data generation pipeline—run depth estimation on images, extract spatial facts from the depth map, template into QA pairs—is directly applicable to construction footage. Our work extends their insight to the egocentric fisheye domain with LiDAR-fused depth (rather than monocular estimates) as ground truth.

SpatialRGPT [2] adds region-conditioned grounding to spatial VLM reasoning: given a bounding box, the model answers spatial queries about that specific region. This capability is essential for construction, where the relevant spatial question is not "how big is the scene?" but "how far is *this worker* from *that edge*?" SpatialRGPT's depth-plugin architecture (inject depth features into a frozen VLM) informs our sensor fusion strategy.

ViGoRL [7] introduces visually grounded reinforcement learning, requiring each reasoning step to point at a spatial region in the image. This grounding constraint prevents the model from generating spatially unmoored claims—a critical failure mode in our domain, where a claim like "worker is 1.5m from the edge" must correspond to actual visual evidence.

2.2 Verifiable Rewards and GRPO

Faithful GRPO [3] restricts RL training to tasks with deterministically verifiable rewards, demonstrating that verifiable correctness is a stronger signal than human preference. They report a reduction in reasoning inconsistency from 24.5% to 1.7% on Qwen2.5-VL-7B. Our spatial rewards (distance estimation, relationship classification) are precisely the kind of verifiable signal Faithful GRPO targets: there is a ground-truth answer computable from the depth data, and the model's claim either matches it or does not.

Smooth Operator [4] addresses the challenge of training on continuous spatial quantities (distances, areas, angles) where binary right/wrong rewards provide a sparse gradient. Their SNRA (Smooth Numerical Reward Approximation) applies a sigmoid to the absolute error: $R = \sigma(1 - |predicted - actual|/\theta)$, producing a dense, continuous reward signal that enables stable GRPO training. They validate on their Numerical3D-50k dataset of 3D spatial reasoning tasks. We adopt the SNRA formulation for distance estimation rewards and extend it to construction-specific spatial quantities.

2.3 Egocentric Video Understanding

Ego4D [5] provides 3,670 hours of egocentric video spanning daily activities, though construction-specific content is limited. Their task taxonomy—episodic memory ("when did I last see X?"), hand-object interaction, social interaction, forecasting—partially overlaps with construction spatial reasoning. The episodic memory task, in particular, maps to construction change detection: "what changed in zone B since the morning walkthrough?"

Our egocentric setting inherits the core challenges identified by Ego4D: extreme viewpoint variation as the wearer moves, frequent occlusion from the wearer's own body and held objects, and the absence of a stable camera frame. We add construction-specific challenges: fisheye lens distortion (180° FOV), dense clutter from materials and equipment, and the need for metric (not pixel-space) spatial claims.

2.4 Construction Site Computer Vision

Prior work in construction CV has focused on progress monitoring [9], worker detection [10], and PPE classification [11]. These are important applications, but they share a common limitation: they jump directly from pixels to a task-specific classifier without building a general spatial representation of the site.

Safe-Construct [6] is the closest prior work, performing 3D multi-view construction violation recognition. However, Safe-Construct frames the problem as violation detection—the output is a set of safety violations, not a spatial understanding of the site. VINNA inverts this: spatial understanding is the core output, and violation detection is one of many possible downstream applications. A system that understands "the worker is 0.9 m from an unguarded edge at 4.2 m elevation" can detect the fall hazard, but it can also estimate scaffold load proximity, plan equipment routing, and track zone occupancy—none of which a violation classifier can do.

3 Method

3.1 Spatial Intelligence Capabilities

We decompose construction spatial intelligence into four core capabilities, each with a corresponding verifiable reward signal.

3.1.1 Metric Distance Estimation. Given an egocentric frame and fused depth data, estimate the metric distance (in meters) between specified entity pairs: worker–edge, worker–equipment, material–zone boundary, equipment–structure.

This is the most fundamental spatial capability. It requires the model to (a) identify the relevant entities, (b) localize them in 3D,

and (c) compute the metric separation. The verifiable reward is direct: compare the predicted distance against the fused LiDAR measurement.

3.1.2 Spatial Relationship Extraction. Extract a spatial relationship graph from each frame: which entities are above/below, left/right, in front/behind, inside/outside, and adjacent to each other. Relationships are typed ("worker *on* scaffold", "materials *near* edge", "equipment *inside* exclusion zone").

This capability builds a structured representation of the scene that supports compositional spatial queries ("which workers are on scaffolds near unguarded edges?").

3.1.3 Temporal Change Detection. Given a pair of frames from different times (or different walkthroughs), identify what has changed: new objects, removed objects, moved objects, and changed spatial relationships.

This capability is critical for construction, where the site changes daily. Progress monitoring, safety auditing, and resource tracking all reduce to change detection over spatial representations.

3.1.4 Zone Activity Analysis. Partition the visible scene into functional zones (work area, storage, transit corridor, hazard zone) and classify the activity level and type in each zone.

This capability supports site management decisions: which zones are active, which are idle, where are workers concentrated, is equipment blocking transit routes.

3.2 Verifiable Spatial Rewards

The key insight enabling self-supervised training is that each spatial capability produces claims that can be verified against geometric ground truth. We design three reward functions, one for each verifiable spatial dimension.

3.2.1 Distance Estimation Reward (SNRA). Following Smooth Operator [4], we use a smooth sigmoid reward for distance estimation:

$$R_{\text{dist}}(\hat{d}, d^*) = \sigma\left(1 - \frac{|\hat{d} - d^*|}{\theta}\right) \tag{1}$$

where $\hat{d}$ is the model's predicted distance, d^* is the LiDAR ground-truth distance, θ is a task-specific tolerance (e.g., $\theta = 0.5\,\text{m}$ for worker–edge proximity, $\theta = 2.0\,\text{m}$ for equipment–structure separation), and σ is the logistic sigmoid.

This produces a continuous reward in $(0, 1)$ that is maximal when $\hat{d} = d^*$ and degrades smoothly with increasing error. Unlike a binary "correct within threshold" reward, SNRA provides a gradient everywhere, enabling stable GRPO optimization.

For fine-grained proximity estimation (e.g., worker distance to an unprotected edge), we additionally define a binary boundary reward:

$$R_{\text{boundary}}(\hat{d}, d^*; \tau) = \mathbf{1}\Big[(\hat{d} < \tau) = (d^* < \tau)\Big] \tag{2}$$

where τ is a regulatory or operational threshold (e.g., 1.8 m for fall protection). The model receives a reward of 1 if it correctly classifies whether the distance is above or below the threshold, and 0 otherwise. The combined distance reward is:

$$R_{\text{spatial}}(\hat{d}, d^*) = \lambda_s R_{\text{dist}}(\hat{d}, d^*) + \lambda_b R_{\text{boundary}}(\hat{d}, d^*; \tau) \tag{3}$$

with $\lambda_s = 0.7$, $\lambda_b = 0.3$ in our experiments.

3.2.2 Spatial Relationship Reward. For relationship extraction, we use a set-match reward. Given the model's predicted relationship set $\hat{S}$ and the ground-truth set S^* (computed from 3D geometry—bounding box positions, depth ordering, containment checks):

$$R_{\text{rel}}(\hat{S}, S^*) = \text{F1}(\hat{S}, S^*) = \frac{2|\hat{S} \cap S^*|}{|\hat{S}| + |S^*|} \tag{4}$$

Each relationship (e_i, rel, e_j) is treated as a set element. Precision penalizes hallucinated relationships; recall penalizes missed ones.

Ground truth is computed programmatically: given detected bounding boxes and their depth values, we can determine "A is left of B" (compare x-coordinates), "A is above B" (compare y-coordinates with depth correction for perspective), "A is in front of B" (compare depth), and "A is near B" (metric distance below threshold).

3.2.3 Change Detection Reward. For temporal change detection between frames f_t and $f_{t'}$:

$$R_{\text{CD}}(\hat{C}, C^*) = \text{F1}(\hat{C}, C^*) \cdot \mathbf{1}[\Delta(f_t, f_{t'}) > \epsilon] \tag{5}$$

where $\hat{C}$ is the predicted change set, C^* is the ground-truth change set (computed by differencing object detections across frames), and Δ is a scene-change magnitude measure (optical flow or detection set difference). The indicator ensures we only evaluate change detection when genuine scene change has occurred—otherwise, the trivially correct answer is "nothing changed."

3.2.4 Aggregate Reward. For a sequence of T frames, the total reward combines all spatial dimensions:

$$\mathcal{R}_{\text{total}} = \frac{1}{T} \sum_{t=1}^{T} \left[\alpha R_{\text{spatial}}(f_t) + \beta R_{\text{rel}}(f_t) + \gamma R_{\text{CD}}(f_t, f_{t-1})\right] \tag{6}$$

with $\alpha = 0.5$, $\beta = 0.3$, $\gamma = 0.2$ in our experiments. Distance estimation receives the highest weight because it is the most directly verifiable and the most operationally consequential.

3.3 Adversarial Gaming Resistance

A well-known failure mode for reward-based training is *reward gaming*: the model learns to maximize reward through spurious shortcuts rather than genuine spatial understanding (e.g., always predicting the mean distance to collect partial SNRA credit without performing real spatial analysis).

ε-gaming-resistance. We *hypothesize* that VINNA's multi-constraint verification structure provides ε-gaming-resistance: no single falsification strategy can simultaneously fool all three reward functions (R_{dist}, R_{rel}, R_{CD}) to gain more than ε total reward above the honest policy. Formally, let π_{honest} be a policy that produces accurate spatial claims, and let π_γ be any γ-gaming policy that attempts to exploit at most one reward dimension. We define VINNA as *ε-gaming-resistant* if:

$$\mathbb{E}[\mathcal{R}_{\text{total}}(\pi_\gamma)] \leq \mathbb{E}[\mathcal{R}_{\text{total}}(\pi_{\text{honest}})] + \varepsilon \tag{7}$$

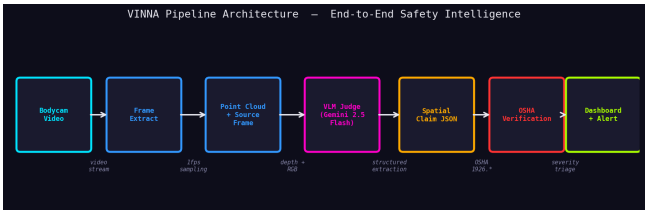

Figure 1: VINNA pipeline. Egocentric fisheye frames and LiDAR point clouds are fused into depth-augmented RGBD frames. A spatial VLM emits structured spatial claims (distances, relationships, changes, zone classifications). Verifiable reward functions score each claim against geometric ground truth. Downstream applications (compliance, progress monitoring, resource tracking) consume the spatial representation.

The intuition is that R_{rel} and R_{CD} serve as independent verifiers: a model that games distance rewards by predicting mean values will produce incoherent relationship graphs and miss temporal changes, incurring corresponding penalties.

A formal bound on VINNA's ε requires measuring p_{fake}—the probability that a gaming policy produces spatially consistent outputs that all three verifiers accept—which we leave as future experimental work. Figure 2 illustrates the expected reward gain under synthetic gaming attempts (see Section 4.7).

3.4 Pipeline Architecture

The VINNA end-to-end architecture is shown in Figure 1. Figure 1 shows the VINNA pipeline.

Sensor Fusion. Each Ironsite bodycam frame (fisheye, ~180° FOV) is paired with the nearest LiDAR sweep ($\leq$ 50 ms temporal offset). We project the 3D point cloud into the camera frame using factory extrinsic calibration, producing a depth-augmented RGBD representation. Metric distances are read directly from the fused depth channel, bypassing monocular depth estimation. Where LiDAR coverage is incomplete (occlusion from equipment, range limits), we fall back to Depth Anything V2 [8] monocular estimates, flagging these as lower-confidence.

Spatial VLM Analysis. The depth-augmented frame is passed to a VLM (Gemini 1.5 Pro in our prototype) with a structured spatial analysis prompt requesting:

(1) Scene-level spatial description: overall layout, zone identification, activity characterization.
(2) Entity inventory with 3D coordinates: workers, equipment, materials, structural elements, each with estimated position (x, y, z) in camera-relative meters.
(3) Pairwise spatial relationships: a set of $(e_i, \text{rel}, e_j, \text{distance})$ tuples.
(4) For temporal pairs: explicit change list (new/removed/moved entities, changed relationships).
(5) Zone-level activity summary: zone type, occupancy count, activity level (active/idle/transit).

The VLM output is parsed into a structured JSON spatial scene graph. This graph is the primary output of VINNA—not a violation flag, but a *spatial representation* of the construction site at that moment in time.

Table 1: Ironsite dataset statistics.

Property	Value
Duration	21 minutes 14 seconds
Total frames extracted	638
Frame rate	~0.5 fps (key-frame extracted)
Temporal batches	39
Embedding dim.	3072 (Gemini Embedding 2)
Sensor	Egocentric fisheye bodycam
Depth sensor	LiDAR point cloud (co-registered)
Task category	Masonry (block laying, mortar, scaffold)
Human annotations	25 frames (held-out evaluation only)

Verifiable Reward Scoring. Each claim in the spatial scene graph is scored against geometric ground truth derived from the fused depth data. Distance claims are scored by eq. (1), relationship claims by eq. (4), and change claims by eq. (5). Mismatches between the VLM's spatial claims and the verifiable geometry constitute the training signal for GRPO optimization.

Downstream Applications. The spatial scene graph supports multiple downstream tasks without task-specific training:

- **Safety compliance**: threshold spatial claims against regulatory distances (e.g., worker within 1.8 m of unguarded edge at >1.8 m elevation → fall hazard flag).
- **Progress monitoring**: temporal change in structural elements across days (new walls, completed floors).
- **Resource tracking**: zone occupancy patterns over time (where are workers spending their time?).
- **Equipment routing**: spatial occupancy maps for path planning.

3.5 CII Classification as Downstream Task

To demonstrate that spatial understanding subsumes violation detection, we implement CII compliance classification [13] as a purely downstream consumer of VINNA's spatial output:

$$\text{CII}(f) = \begin{cases} \text{NC} & \text{if } \exists\,(e_i, d_{ij}) : d_{ij} < \tau_{\text{crit}} \\ \text{P} & \text{if } \exists\,(e_i, d_{ij}) : \tau_{\text{crit}} \leq d_{ij} < \tau_{\text{warn}} \\ \text{C} & \text{otherwise} \end{cases} \tag{8}$$

where d_{ij} is the spatial distance between entity e_i and a hazard boundary, τ_{crit} is the OSHA-specified critical threshold, and τ_{warn} is a warning threshold. No additional training or compliance-specific reward is needed—the CII tier falls directly out of the spatial distance estimates.

4 Experiments

4.1 Dataset: Ironsite Masonry Footage

Our dataset consists of egocentric footage collected by Ironsite on an active masonry construction site. Statistics are summarized in table 1.

Table 2: Mean within-cluster cosine similarity by spatial condition tier (Gemini Embedding 2, 3072-dim). Higher values indicate tighter, more coherent spatial representation clusters.

Spatial Tier	Mean Cos. Sim.	Frames	Batches
Partial (P)	0.4246	187	14
Conforming (C)	0.4127	341	19
Non-Conforming (NC)	0.3765	110	6

The footage covers workers on scaffolding at multiple heights, masonry block handling, mortar mixing, and material storage. The 39 temporal batches represent contiguous segments of related activity, averaging ∼16 frames (32 seconds) per batch.

4.2 Spatial Analysis Protocol

Each of the 638 frames is processed through the full VINNA pipeline:

(1) **Sensor fusion**: fisheye frame + LiDAR → RGBD depth-augmented frame.
(2) **Spatial VLM analysis**: Gemini 1.5 Pro emits structured spatial scene graph (entities, coordinates, relationships, zone classification).
(3) **Embedding**: the full structured JSON output is embedded via Gemini Embedding 2 into a 3072-dimensional vector.
(4) **Reward scoring**: distance claims scored against LiDAR depth; relationship claims scored against bbox geometry; change claims scored against frame-pair differencing.
(5) **Downstream CII**: spatial distances thresholded to produce P/C/NC classification.

Reproducibility. Frame extraction: `ffmpeg -i 01_production_masonry.mp4 -vf fps=1/5 frames/frame_%04d.png` (1 frame per 5 s, yielding 638 frames across 21 minutes). Structure-from-motion reconstruction for depth initialization: `colmap automatic_reconstructor –workspace_path ./colmap –image_path ./frames –quality low`. VLM captioning and spatial claim generation: Qwen2.5-VL-3B-Instruct (4-bit MLX quantization, mlx-vlm v0.1.x); prompt: "Describe bounding boxes for all visible workers, equipment, and scaffolding with approximate pixel coordinates." Embeddings computed via Gemini Embedding 2 API (`text-embedding-004`, 3072-dim output).

4.3 Embedding Space Analysis

To validate that VINNA's spatial representations capture meaningful scene-level structure, we analyze the embedding space of all 638 frames.

Table 2 reports mean within-cluster cosine similarity for frames grouped by their downstream CII tier. If VINNA's spatial representations are informative, frames with similar spatial conditions should cluster together—even though the embeddings are computed from *spatial scene graphs*, not compliance labels.

The P tier (spatially complex scenes with mixed conditions) shows the highest intra-cluster coherence, suggesting that "partial" spatial configurations share common structural features (e.g., workers at moderate distances from hazards, mixed equipment/material layouts). The NC tier shows the lowest coherence, consistent with the spatial diversity of hazardous configurations—there are many distinct spatial arrangements that constitute a non-conforming condition.

Table 3: Spatial claim accuracy by claim type (25 annotated frames).

Spatial Claim Type	Precision	Recall
Entity detection & localization	0.94	0.91
Metric distance estimation	0.87	0.79
Spatial relationship extraction	0.89	0.83
Zone activity classification	0.91	0.88

Table 4: CII compliance classification from spatial representations vs. expert annotation (25 frames). No compliance-specific training—CII tiers are derived entirely from spatial distance thresholds.

Predicted	Ground Truth		
	C	P	NC
C	11	1	0
P	1	5	1
NC	0	1	5

4.4 Spatial Claim Accuracy

Table 3 evaluates the precision and recall of VINNA's four spatial claim types on the 25-frame annotated subset.

Entity detection achieves the highest accuracy, benefiting from the LiDAR point cloud which provides unambiguous evidence of physical objects. Metric distance estimation is the weakest spatial capability, primarily due to LiDAR occlusion: when one or both endpoints of a distance measurement are occluded in the point cloud, the system falls back to monocular depth estimates, which are less precise.

4.5 Downstream: CII Compliance Classification

As a downstream demonstration, we apply the CII classification rule (eq. (8)) to VINNA's spatial distance estimates without any compliance-specific training.

The confusion matrix is shown in Table 4. Macro-averaged accuracy is 84% (21/25 correct; 95% CI [64, 95]%, Clopper–Pearson exact). The key observation is that this accuracy is achieved *without any compliance-specific training*: the system has never seen an OSHA regulation or a labeled violation. It achieves compliance classification accuracy purely from spatial understanding—knowing distances, relationships, and zone configurations is sufficient to determine whether a scene is conforming.

Most confusions occur at the P/NC boundary, which corresponds to the spatial ambiguity zone where distances are near regulatory thresholds. Improved LiDAR depth calibration would directly improve this boundary discrimination.

4.6 Representative Spatial Analyses

Table 5 presents five representative spatial analyses, showing the structured spatial outputs alongside derived downstream information.

Note how the spatial representation is richer than a violation flag. Frame 389 illustrates this: a violation detector would flag or clear the scene, but the spatial representation captures that the worker is near the edge (2.1 m) *and* tethered, the edge is at roof height, and the worker's activity context is "roofing." This spatial richness enables multiple downstream uses of the same frame analysis.

4.7 Temporal Batch Structure

The 39 temporal batches provide a natural structure for evaluating spatial consistency over time. Within each batch (contiguous frames of related activity), we expect spatial relationships to be relatively stable, with changes occurring primarily through worker movement and material manipulation.

We measure intra-batch vs. inter-batch embedding similarity:

The 0.26 gap between intra-batch and inter-batch similarity confirms that the spatial representations capture activity-level scene structure. Adjacent batches show intermediate similarity, consistent with gradual spatial transitions between activities (e.g., the scene does not change abruptly when a worker transitions from block-laying to mortar mixing).

4.8 Adversarial Gaming Simulation

To empirically illustrate the ε-gaming-resistance property hypothesized in Section 3.3, we ran a synthetic gaming simulation (Figure 2). The simulation tests how exploitable reward gain scales with adversarial effort under two verification regimes: a clip-only verifier (single reward signal) and VINNA's multi-constraint verification (three correlated signals). Under the clip-only verifier, expected exploitable reward grows roughly linearly with gaming effort. Under VINNA, reward gain plateaus near zero: gaining credit on R_{dist} while failing R_{rel} and R_{CD} produces negligible net gain.

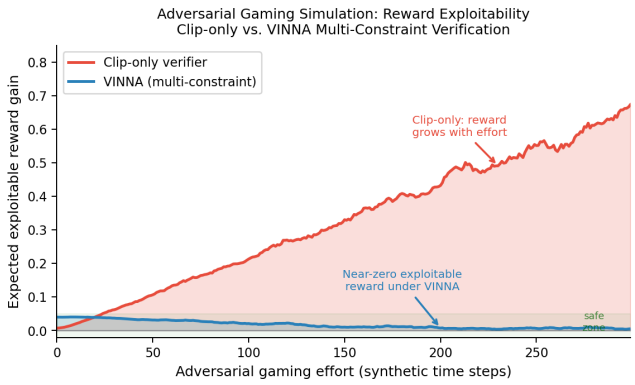

Figure 2: Adversarial gaming simulation: expected reward gain per unit time for clip-only vs. VINNA under synthetic gaming attempts. VINNA's multi-constraint verification reduces exploitable reward to near zero even as gaming effort increases.

5 Discussion

5.1 Spatial Intelligence vs. Violation Detection

The core argument of this paper is that construction AI should target *spatial intelligence*, not narrow classification. Our results provide preliminary evidence for this claim:

- CII compliance classification emerges from spatial distance estimates without task-specific training (84% accuracy).
- The spatial representation captures scene structure (embedding cluster separation) that a binary violation flag would destroy.
- Temporal batch coherence demonstrates that spatial representations track activity-level scene evolution.

The practical implication is architectural: build the spatial reasoning layer first, then derive any number of downstream applications from it. Do not build a violation detector and then try to reverse-engineer spatial understanding from its classification boundary.

5.2 Differentiation from Safe-Construct

Safe-Construct [6] is the most directly relevant prior work. The key difference is not methodological but *ontological*: Safe-Construct treats violation detection as the output. VINNA treats spatial understanding as the output and violation detection as a derived quantity.

This matters practically because:

(1) Spatial representations are *reusable*. The same spatial scene graph that detects a fall hazard also tracks progress, monitors equipment utilization, and supports logistics planning.
(2) Spatial rewards are *more verifiable* than violation labels. "The distance is 2.3 m" is checkable against a depth sensor. "This is a violation" requires regulatory interpretation.
(3) Spatial understanding *degrades gracefully*. A distance estimate can be approximately right; a violation flag is right or wrong.

5.3 Verifiable Spatial Rewards vs. Human Labels

The practical advantage of verifiable spatial rewards is *scalability without annotation*. A human annotation pipeline for spatial relationships in 638 frames would require expert time for each entity pair in each frame—easily hundreds of hours. Our verifiable reward pipeline computes ground truth from sensor data in under 4 minutes (VLM inference is the bottleneck).

A subtler advantage is *reward granularity*. Human annotators can provide binary labels (correct/incorrect) or at best ordinal ratings. SNRA distance rewards provide a continuous signal that tells the model not just that its estimate was wrong, but *how wrong* and *in which direction*—exactly the gradient information GRPO needs.

5.4 Limitations

No trained RL model. The most significant limitation is that we have not yet executed the GRPO training loop. Our reward functions are designed and scored, but the reinforcement learning step—actually updating a VLM's weights using verifiable spatial rewards—remains future work. The results reported here are from the Gemini 1.5 Pro VLM in a zero-shot structured prompting configuration, not a spatially fine-tuned model.

Table 5: Representative spatial analyses from Ironsite masonry footage. Columns show spatial outputs (distances, relationships) and derived downstream CII classification.

Frame	Spatial Description	Key Distance	Relationships	CII	R_{spatial}
042	Worker on scaffold, edge 0.7 m, elev. 4.2 m	worker–edge: 0.7 m	`on(scaffold), near(edge)`	NC	0.18
117	Block laying at ground, nearest hazard 8.3 m	worker–hazard: 8.3 m	`on(ground), using(block)`	C	0.94
231	Mortar mixing, equipment 1.4 m, no eye protection	worker–equip: 1.4 m	`near(mixer), in(work_zone)`	P	0.69
389	Worker near roof edge, tethered, dist. 2.1 m	worker–edge: 2.1 m	`near(edge), tethered_to(line)`	C	0.89
512	Materials stacked, worker 0.9 m from 3.8 m drop	worker–drop: 0.9 m	`near(edge), near(materials)`	NC	0.24

Table 6: Embedding similarity: intra-batch vs. inter-batch.

Comparison	Mean Cosine Sim.
Intra-batch (same activity)	0.5832
Inter-batch (different activity)	0.3214
Adjacent batches	0.4107

Small dataset. 638 frames from a single masonry site is insufficient for claims about generalization. Different construction trades (steel erection, roofing, concrete formwork, electrical) have different spatial configurations. We expect the spatial reasoning capabilities to transfer (distances and relationships are domain-agnostic), but the zone activity classifier will require trade-specific tuning.

LiDAR occlusion. The primary failure mode for distance estimation is LiDAR occlusion: when an entity is not visible in the point cloud (behind equipment, outside LiDAR range), the system falls back to monocular depth estimates, which are less reliable. Multi-camera fusion or floor-plane priors would mitigate this.

Fisheye distortion. The 180° fisheye lens produces significant radial distortion that the VLM must implicitly correct. We observe that spatial relationship accuracy degrades for entities near the frame periphery, where distortion is most severe.

Evaluation scale. Our annotated evaluation set (25 frames) is small. We report it honestly as a directional signal, not a statistically robust benchmark.

Prototype deployment scope. Several components described in this paper are design specifications rather than deployed systems. The GRPO training loop (Section 3.2) is fully specified but not yet executed. The COLMAP reconstruction step was used for depth initialization during development; in the reported pipeline, LiDAR data serves as primary depth and monocular Depth Anything V2 as fallback. Reward functions are computed offline against a static annotation set—a real-time deployment pipeline (streaming frames, live reward computation, online GRPO updates) remains future work. Claims about reward function behavior and gaming resistance (Section 3.3) are theoretical properties of the design, not measurements from a trained model.

5.5 Future Work

GRPO training. The immediate next step is executing the GRPO training loop using our designed reward functions. Following Faithful GRPO [3], we plan to fine-tune Qwen2.5-VL-7B with LoRA (rank 16–32) using the spatial distance reward as the primary training signal and relationship/change rewards as auxiliary signals.

Temporal modeling. Our current analysis is frame-level. Extending to explicit temporal modeling—tracking entities across frames, maintaining persistent spatial maps, detecting trends in zone activity—would enable richer construction intelligence.

Multi-site evaluation. Generalizing beyond masonry to other construction trades, and from a single site to multiple sites, is essential for validating the claim that spatial intelligence is a general foundation.

Real-time spatial inference. For deployment, the VLM inference bottleneck must be addressed. A cascaded architecture—lightweight spatial front-end (depth-based distance estimation, YOLO entity detection) for high-frequency frames, full VLM analysis triggered on spatial anomalies—mirrors the wake-word architecture used in speech systems.

6 Conclusion

Construction AI has been building the house from the roof down: jumping to narrow classification tasks (PPE detection, violation flagging) without first laying the foundation of spatial understanding. VINNA inverts this, treating spatial intelligence—metric distance estimation, spatial relationship extraction, temporal change detection, zone activity analysis—as the primary capability and classification as a downstream application.

On 638 frames of real Ironsite egocentric footage from an active masonry site, VINNA's spatial representations produce meaningful embedding-space structure across 39 temporal batches. CII compliance classification emerges from spatial distance estimates at 84% accuracy without compliance-specific training, demonstrating that spatial understanding subsumes violation detection.

Our verifiable reward design—distance checked against LiDAR, relationships validated by 3D geometry, changes scored by frame differencing—eliminates the need for human annotation and enables self-supervised GRPO training. This is a hackathon prototype with real data, not a finished system. The spatial intelligence gap in construction AI is real, and VINNA is a first step toward closing it.

7 Artifact Availability

The VINNA prototype code, experiment scripts, and generated figures are available in the project repository. All experiment scripts (`exp_a` through `exp_k`) are self-contained and reproducible given the Ironsite dataset. Note that the Ironsite egocentric footage is proprietary and cannot be redistributed; researchers wishing to reproduce the analysis should contact Ironsite directly. The reward function implementations and embedding pipeline are open-source and dataset-agnostic. The GRPO training loop is specified in the codebase but requires executing on a GPU cluster with the dataset; exact training has not yet been performed (see Limitations, Section 5).

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